



Volatility Weighted Momentum

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1 Main Idea

The strategy captures medium-term trends in futures markets by calculating momentum over fixed lookback windows (e.g., 20, 50, 100 days), with potential for dynamic horizon selection in future iterations. These signals are normalized by volatility forecasts—using methods such as rolling standard deviation, GARCH(1,1), EWMA, or downside deviation—to generate standardized, cross-asset scores. Position sizes are scaled according to these risk-adjusted momentum scores, ensuring comparability across markets and alignment of exposure with opportunity strength.

2 Economic Thesis

The underlying economic thesis of this strategy is that markets exhibit persistent, directional trends due to behavioral and structural frictions, but these trends are not constant in strength or reliability. Momentum exists because investors tend to underreact to new information, herd into established trades, and face institutional or regulatory constraints that slow the incorporation of fundamentals into prices. These frictions lead to return persistence over medium horizons, making momentum a robust and empirically supported source of excess returns across asset classes.

However, returns from momentum are highly sensitive to volatility conditions. Periods of elevated volatility often coincide with regime shifts, forced deleveraging, or liquidity shocks, during which trends can rapidly break down. By scaling momentum signals with volatility estimates—such as GARCH forecasts, risk-parity-type dynamic scaling, or downside-sensitive measures—the strategy accounts for the fact that risk premia are time-varying and that investors require higher compensation to bear risk in turbulent environments.

The use of volatility-adjusted momentum also facilitates cross-asset comparison, enabling capital to be deployed more efficiently across diverse futures markets. Finally, the incorporation of regime-switching logic reflects the observation that trend-following performs best during stable, directional markets but is challenged during mean-reverting or policy-driven periods. Adjusting exposure according to regime improves robustness and aligns the strategy with broader macroeconomic and liquidity cycles.

3 Implementation

The strategy is implemented by calculating various rolling momentums (20, 50, 100 days) for all futures data we currently have access to. We also calculate various rolling volatility measures including typical rolling standard deviation (20 and 50 days), GARCH(1,1), EWMA(0.94), and downside deviation. The score which determines position entry and sizing is determined by dividing a chosen momentum measure by the chosen volatility measure (we are still in the process of testing which ones to ultimately use). We ultimately will compare the performance of each combination of momentum and volatility.

The strategy is implemented by calculating a momentum of 20 days for all futures data we currently have access to. The momentum is then smoothed using a Kalman filter. We also calculate rolling volatility using ___ as well as the rolling second order volatility (volatility of volatility) by taking the rolling standard deviation of the same window for which we are calculating the first order volatility. Then, for any given day, a score is calculated by dividing the momentum by the volatility plus the second order volatility scaled by two. Once all the scores have been calculated for each individual asset for a day, it is fed into a softmax normalization function mapping each asset to a number between 0 and 1 such that the total for all the assets on a day is equal to 1 (in other words, a discrete probability distribution). These results are used to determine the ideal position for an asset on a given day by buying or selling contracts to obtain as close to an allocation as possible to the score. Note that allocations of less than 1% are not taken into account and are in essence looked at as unallocated.

4 Risk Management & Controls

One of our biggest risks is choppy markets causing allocations to fluctuate greatly on a day to day basis which is why we spent the most time addressing that risk. Using Kalman filtering on the momentum allows us to obtain a more smooth measure of momentum. Along with this,

5 Performance

To come after backtesting

6 Appendix

6.1 Supplemental Charts & Tables

6.3 References & Data Sources

Notes for improvements:

- Residual momentum (regress futures against factors such as index, carry, etc) and see how much momentum is isolated
- Also take into account volatility of volatility (avoids kurtosis risk)
- Kalman filter momentum
- How will we define entry/exit sizing (absolute sum, softmax, power normalization)

Loop through each available trading day

Look at momentum calculation, volatility calculation, and vol of volatility calculation

Some sort of measure to combine these 3 into a single score ($\text{mom}/(\text{vol}+a*\text{vvol} + \text{constant})$)

Construct portfolio for the day based on softmax normalization of scores

Use these obtained percentages to define rebalancing of portfolio every day

Record P/L daily